

Chapter 10

Wearable Devices and IoT



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Abstract Wearable devices are the significant ubiquitous technology of the Internet of Things in day-to-day life. The efficient data processing in various devices such as smart clothes, smart wristwear and medical wearables along with consumer-oriented service of the IoT technology becomes inevitable in smart healthcare systems. The wearable market is currently dominated by health, safety, interaction, tracker, identity, fitness etc. Wearables increase the convergence of physical and digital world which automatically bring people into the IoT. The popularity of wearable devices is growing exponentially since it entirely changes the way how the consumers interact with the environment. 74% people believe that the wearable sensors assist them in interacting with the physical objects around them. Henceforth, one out of three smartphone users will wear minimum 5 wearables in 2020. Moreover, 60% believe that wearables in the next five years will be used not only to track health related information, although it can be used to control objects, unlock doors, authenticate identity and transactions. Wearables must be evolved to cope with the future to meet the expectations of consumers, where the users will wear many devices that is connected with the internet to interact with the physical surroundings and receive data in a seamless secure way. By 2021, smartwatches are estimated to be sold to nearly 81

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million units which signifies 16% sales of total wearable device. According to the latest figure of Gartner report, the global shipment of wearable devices are anticipated to raise by 25.8% every year to \$225 million (GBP 176.3 million) in 2019. Researchers also forecasted that the usage of wearable devices by the end users will increase to \$42 billion (GBP 32.9 million) in 2019. In recent years, the IoT based Smart Healthcare system has influenced greatly on growing demand of wearable devices. In fact, the Wearable IoT (WIoT) devices are generating huge volume of personal health data. Enabling technologies such as cloud computing, Fog computing and Big Data play vital role in leveraging WIoT services. These enabling services over the voluminous health data enhance clinical process at health care system at remote or local servers. The traditional remote healthcare information system involves data transfer, signal processing mechanism and naive machine learning models deployed on remote server to process the medical data of patients. This technique has several demerits like they are not suitable for resource constrained wearable IoT devices. The resources such as processing, memory, energy, networking capability are limited in WIoT devices. Traditional mechanism lacks optimization of resource usage, prediction of medical condition, and dynamic assessment based on available information. Further, the naive machine learning techniques does not perform knowledge generation, decision making and discover hidden valuable patterns from the available medical data. The integrated platform in which cloud computing serves as backend computing systems, Fog computing as edge computing and Big data as platform for data analysis, knowledge generation promise to provide valid solution to several issues of Wearable IoT devices. Next, the health data generated through WIoT devices are personal and sensitive. Hence, the security and privacy of such delicate data at all level of WIoT ecosystem is essential. This part of chapter will contribute towards understanding the recent research work, issues, challenges, and opportunities in applying enabling technologies for WIoT. Also, how well the security and privacy can be incorporated is also discussed.

Keywords Wearable devices · Internet of things · Body area network · Healthcare · Smart devices

10.1 Introduction

Wearables enhance the quality of human life that cannot be achieved with the smartphone alone and the users are showing an incredible interest towards in wearing these devices practically. Smart watches, smart jewellery, smart glasses, skin patches, electronic garments and so on, are commonly referred to as wearables. Wearable devices can sense, gather, store physiological data and improves the human life quality which is not possible with smartphones alone. Wearables perform multiple micro-level tasks such as verifying incoming messages and immediate processing of urgent data more naturally and conveniently. Wearables are capable of providing set of value-added services such as physical and mental health monitoring [1, 2], indoor localization

and navigation [3, 4], medical insurance analytics [5], sport analytics [6], and financial payments [7, 8]. According to recent market statistical report [9, 10], the yearly shipment is anticipated to reach 200 million by 2019. The market value of wearable technology is predicted to reach \$57,653 million by 2022, which is nearly \$19,633 million that is 3 times more than in 2016 [11].

Wireless Body Area Network (WBAN) has also become very popular in the health monitoring aspect and the vision of WBAN is to have a wider network with the collection of wearable sensors and actuators. It comprises medically implanted devices which provide accurate and comprehensive information about the human health. Indeed, WBAN needs persistent research efforts in various layers such as physical, Medium Access Control (MAC) and network. Moreover, the significant factors such as independent usage, low-cost increase the utilization of wearables without any significant intervention of medical professionals and it makes opportunity to the wide deployment of wearable devices.

The main objective of patient-centric system is to improve patient health, experience with low cost [12]. The system relies on large data sets to find out clinically relevant sub categories and disease mechanisms. Therefore, high quality measurement is required to construct high quality information content data. The physiological measurement, cellular and molecular profiles plays a significant role in determining individual's health state. Wearable technologies advance precision medicine system by defining measurement of clinically relevant parameters, high quality information content.

Wearable technology enables continuous digital measurement of parameters about individual's health state in real time. It is observed as a major component in fitness market and today's lifestyle with photoplethysmography-based heart monitors and accelerometer-based activity monitors amongst the huge amount of available devices. A significant challenge in the utilization of wearables in medicine is that only few devices are approved for the use in healthcare system. Nevertheless, more opportunities are there for developing and adapting wearable devices for healthcare use. In healthcare system, the four primary important measurements such as heart rate, temperature, blood pressure, and respiration rate are often limited in a physical visit to a primary care physician. Continuous measurements can be made but with limited portability, but the wearable devices enable longitudinal measurement of these devices with high precision. The key principle in the deployment of wearable devices in the healthcare system obtains high quality, clinically relevant data to physicians and patients. The measurement of bio-signals such as heart rate, temperature, blood pressure, respiration rate, physical activity, sweat and emotion are reviewed [13–15].

Recently, there is an exponential growth in the huge variety of wearables in the market which provides different wearing options and functionalities. Several research challenges and solutions related to wearables with the current trends and technologies are discussed. The research directions and future market strategies for next generation wearable devices in the perspective of design and development are also discussed.

10.1.1 *Internet of Things*

The definition of IoT provided by U.S. National Intelligence Council [16] “It is the general idea of physical things, i.e. everyday objects, which are readable, identifiable, locatable, controllable and addressable with the internet via wireless LAN, RFID, Wide Area Network. Particularly, everyday objects are equipped with sensors and microprocessors, can be communicated with each other. It includes different items such as household appliances, clothing, food, commodities, materials, landmarks, luxury items, buildings, roads, etc.

It is predicted that microprocessors are embedded with 5% of objects [17] and the small microprocessor chips and sensors read and transmit data such as temperature, sound waves, movement etc. According to the report released by Cisco, it is estimated that 50 billion connected devices by 2020, seven times the world’s population [18]. GSMA predicts that the count of total internet connected devices would be 24 billion by 2020 [19]. The three significant categories of IoT featured by market segment are: automotive and transportation applications, tracking and controlling activities in homes and buildings, health and personal environment monitoring.

IoT technology enable automation from small-scale to large-scale industries and it plays a significant role in fourth industrial digital revolution. It has found a standard market place for various commercial uses such as precision agriculture, water usage management, smart parking etc. An extensive research is being conducted for developing intelligent systems in the areas such as smart grids, crash avoiding cars, structural health monitoring, traffic congestion minimization. The data collection, reporting and remote monitoring of different objects have proven in aforementioned domains. It can be further expanded for monitoring patient’s health and intimate to doctors, caretakers, healthcare centers and emergency services whenever required.

IoT is an emerging trend and has a huge impact on several aspects of life such as industry, environment, entertainment, sports, social computing, food processing, agriculture, education, defects, city lives, astrophysics etc. Nowadays, human lives depend on the e-healthcare service and IoT plays a significant role in the healthcare system. Researchers involve in the invention of intensive approaches for the development of novel techniques on e-healthcare. According to a recent statistical report, the adoption of IoT may bring incredible changes in the operational efficiency to surgical centers and hospitals in managing day-to-day clinical operations and tracking the health status of patients. Cost efficiency, safety and reliability are the significant factors achieved with IoT applications in the e-healthcare system. The global IoT e-healthcare market is predicted to increase from \$32.4 Billion in 2015 to \$163.2 Billion by 2020. In particular, the connectivity between wearables and IoT empowered technologies is significant matter of study, where the interaction means how the user gets connected with the system (e.g. 2G, 3G, 4G, Near-field communication (NFC), Wi-Fi (Wireless Fidelity), Bluetooth Low Energy (BTLE)). The “Things” represents objects such as wearable sensor devices that senses the environment, collects data and it is processed.

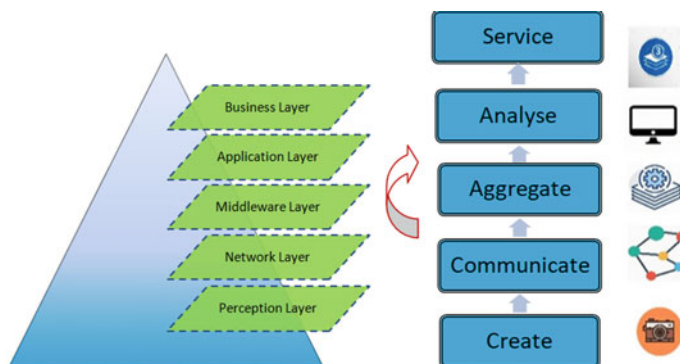


Fig. 10.1 IoT architecture

The Internet of Things (IoT) facilitate human lives by using many devices such as smartphones, PCs, tablets, laptop, server and the devices connected to the internet. There is a rapid growth in the number of people using connected devices every year. IoT connects physical things to the internet and transfers information about the surroundings. IoT affords solutions by integrating technologies such as hardware and software to store, retrieve, process and communicate the data among individuals or groups. The adoption of IoT in healthcare has ended up in a wide opportunity because of its significant factors such as cost-effective, smart and more accurate healthcare services. The inevitability of self-health monitoring is rapidly growing due to the tremendous increase in the amount of elderly people until 2020. The IoT based wearable devices connected to the internet using Bluetooth or Wi-Fi can be worn as accessories or can be attached to the clothes. The wearable devices sense, analyze, store, transmit and utilize the data based on the application.

The healthcare IoT architecture consists of three layers: perception layer, network layer, application layer. The paradigm assists clinicians to monitor patient's health remotely and afford timely assistance in all emergency situations [20]. Figure 10.1 show the layers in IoT architecture which are extended to five and described below:

Perception Layer: It includes several kinds of sensors to identify the physical condition and location of the user. It is also known as “Device Layer” consists of several physical objects and sensor devices. The sensor devices such as infrared, RFID, 2D-barcode locate, sense and collects information about the object. It can also be identified by sensing other physical parameters such as temperature, vibration, motion, location, acceleration, humid level, chemical changes etc. The information is then passed to the network layer for further processing.

Network Layer: It is also known as “Transmission Layer” and it connects several network devices, smart objects, servers. The data gathered from the sensor devices is safely transmitted and processed. The transmission media can be Wi-Fi, infrared, ZigBee, UMTS, 3G, Bluetooth etc. Then the information is transmitted to the processing layer. It contains the network and transport capabilities. The information

gathered from sensors transmitted to the receiver and then transferred to the system. At last, it is transferred to healthcare monitoring center for further processing via internet.

Middleware Layer: It is the processing layer which stores huge amount of data for analyzing and processing. The significant responsibility of this layer is to maintain the connection with the database server and service management. The devices which are providing same set of services can be interconnected with each other.

Application Layer: The main responsibility lies in delivering customer oriented services to the user and directly interacts with the end user. It combines IoT technology and healthcare center to provide health service remotely to the user.

Business Layer: The vital role of the layer is in controlling and managing the complete IoT system including user's privacy, applications, and business models. The different flowcharts, graphs, business models are created based on the data retrieved from the application layer.

The three main concepts are vital to gain a deep insight about wearable ideas and concepts in the industry: Safety and security, smart garments and accessories, non verbal communication.

10.1.2 Internet of Things Healthcare

Remote health monitoring system is plausible since it could more benefits in different contexts. It could provide feasible access for elderly people living independently at home or people who are living in rural areas. In particular, it minimizes the strain for healthcare users such as non-critical patients at home rather staying in hospitals. The remote health monitoring system gives better control for the patients to maintain their health at all times. The remote monitoring healthcare system minimizes the cost and overcrowd of hospital beds.

Relatively, there are few drawbacks in remote health monitoring system include individual sensors need to be recalibrated regularly which ensures accuracy in monitoring process, loss of connectivity if the devices ran out of battery or out of cellular range, security risk arises due to the storage of immense amount of data in a single database. As progress continues to solve such kind of issues and IoT based healthcare systems would provide a viable solution in the near future.

IoT healthcare system have been developed for several purposes include diabetes management, rehabilitation, Assited Ambient Living (AAL) etc. After any physical injury, rehabilitation has been a hot research topic and it is tailored towards an individual focusing on their symptoms.

The current status of patient's health is compared with symptoms, ailments and treatments of previous patients details stored in the database. Doctor manually feeds the symptoms and the recommended treatment is approved. In 87.9% cases, the modifications were not allowed in the proposed treatment plan once the doctor completely agreed with the system. Mathematical models were proposed for measuring

the joint angles in physical hydrotherapy and the improvement in joint movement is also monitored through therapy. IoT enabling technologies assist in monitoring the patients suffered from Parkinson's disease. Wearable sensors function along with vision-based technologies is used to observe tremors, gait patterns, activity levels in the progression of Parkinson's disease. The blood glucose level is monitored in diabetic patients and the readings are noted by the patients at regular intervals in this system. Eventually, there are two types of blood glucose abnormalities. (i) Abnormal blood glucose level (ii) Missed blood glucose reading. Based on the severity of the abnormal condition, the system decides to whom it should be notified; to patient, family members, care takers, emergency healthcare providers such as physicians.

The system can be enhanced by automating the blood glucose measurement process. The system is designed to detect the heart attack with customized antenna and components. An ECG sensor can measure the heart activity and it is processed by the microcontroller. The retrieved ECG data via Bluetooth is processed further and then transmitted to the application layer for presenting data to the user. By measuring the respiratory rate, the heart attack could be predicted with the heart attack prediction software.

10.1.3 Wearable Technologies for Different Body Parts

The working of sensors depends on various techniques like optical, electrical, piezo-electric effect and electrochemical. Different parts of the body have its own type of sensor as shown in figure. The purposes of the wearable sensors are to give valuable information while the users are wearing it. The following section summarizes few wearables which can be worn on body parts and is shown in Fig. 10.2.

10.1.3.1 Head

Head is the uppermost part of the body which includes the eyes, nose, ears, mouth and forehead. The dedicated devices for these parts are Googles, glasses, headbands, contact lenses, earring, hearing aid, patches, earphones and hats. Smart glasses acts as wearable computers embedded with sensors like pressure sensors, gyroscopes, microphones, accelerometers and image sensors. Advanced technologies are used to operate the smart glasses such as voice commands with the help of user interface. JINS MEME, Recon Jet and Google Glass are some of the examples. To operate the Google Glass, the wearers have to rotate their head to thirty degree upward or the touchpad should be tapped with a voice command "OK GLASS". Figure 10.3 shows Google glass teardown.

Smart glasses are used to monitor physiological parameters like respiratory rate and heart rate [21]. Even a Medical introduced wearable glasses to find the patient's vein with the help of sensors [22]. Smart contact lenses are used to improve the vision, detect the glucose levels from tears and monitors the progression of glau-

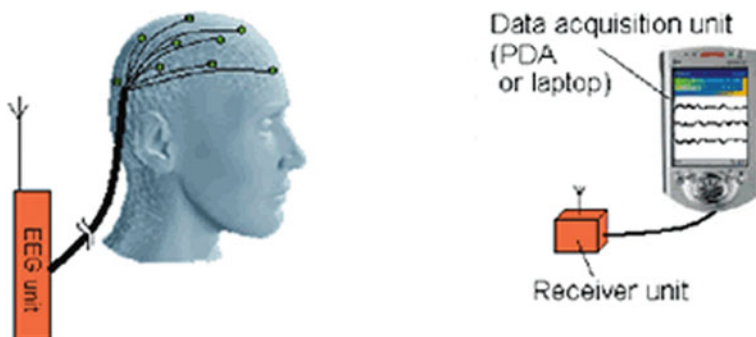


Fig. 10.4 IMEC electroencephalographic (EEG)

coma in patients by measuring the eye lens curvature [23]. Headbands, hats and helmets embedded with wearable sensors monitors the parameters like body temperature, heart rate with the aid of plethysmograph technique. Brain activities can be measured using wearables like SmartCap which captures brainwave signals to identify microsleep that is it last for 5–10 s [24]. The IMEC electroencephalographic (EEG) headset and Melon headband observe the brain's EEG signals to study the mental activity for research and clinical applications [25] and it is shown in Fig. 10.4.

Earbuds, hearing aids, earphones and earrings uses pulse oximetry to capture physiological parameters. X-Patch is a device that can be worn on the back side of the ear to monitor head injuries for industrial workers, sports persons and military personnel [2]. Mouthguard is a wearable device dedicated to be worn inside the mouth to find the various levels of lactate and uric acid in the saliva.

10.1.3.2 Torso

The middle part of the body is torso, which accommodates wearable devices like underwear, belts and suits. Some of the wearable suit includes baby glove swaddles for temperature sensing, smart jackets for monitoring the physiological parameters of newborn infants, military uniforms embedded with computer to monitor and protect military personnels, pajamas for electrocardiogram measurement in babies, swim-suits with a UV sensor and tattoos with wireless connectivity for physical parameter monitors. Genesis is a wearable device combined with smart phones to regulate the blood glucose level [26] and it is shown in Fig. 10.5.

10.1.3.3 Arm, Leg, and Foot

Smart watches, rings, bracelets, wristbands and armbands can be worn to collect physiological parameters like heart rate, blood pressure, UV exposure level, body

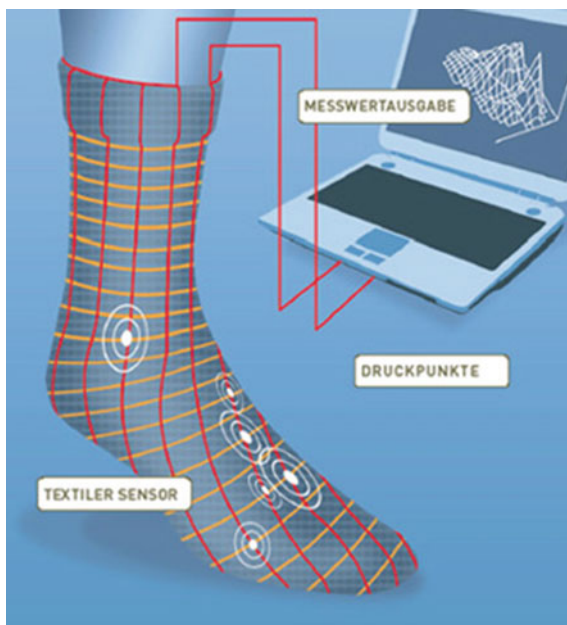
Fig. 10.5 Genesis meter

temperature and daily activities. Smart sleeves and socks are worn on leg or foot used to prevent injury. Alpha-Fit GmbH introduced smart socks which uses 3D technique to measure the pressure distribution over the entire foot when the patient is in motion. This technique is highly appreciated for custom shoe making and the smart socks is shown in Fig. 10.6.

The following graph shows the wearable device unit sales worldwide by region from 2015 to 2017 with a prediction from 2020 to 2022 and it is shown in Fig. 10.7. From the graph it is inferred that the number of wearable devices with 4G connections in North America is amounted to 217 millions. It is predicted that in 2022 it is being doubled and may reach by 439 millions. This drastic growth shows the demand for wearables worldwide.

10.1.3.4 Wearable Devices for Heart Rate

It is a biomechanical organ which pumps blood around the vascular system to provide nutrients and remove waste materials from other organs. The heart rate is dependent on physical activity level such as anxiety [27], sleep [28], illness [29], stress [30] and drugs [31]. Hence, it is a physiological indicator of human health state [32]. Typically, the heart rate at rest is 40–60 bpm for trained human athletes and 60–100 bpm for adults and children more than 10 years. During exercise, the heart rate may reach up to 200 bpm. In common, the heart rate detection depends on the transduction of electrical, mechanical, acoustic or optical signals with the mechanical heart beat rate or from pulsatile hemodynamic flow.

Fig. 10.6 Smart socks

Heart rate can be detected manually by checking the pulsations of an artery near the carotid or radial surface of the body by giving slight pressure with the index and middle fingers. The electrical activity of the heart can be measured by electrodes fixed to the chest i.e. Electrocardiogram (ECG). Pulsatile blood flow can be measured based on variations in 4-electrode resistance method, where electrodes are attached to the neck and waist i.e. Impedance CardioGram (ICG). The stethoscope or ultrasound sensor measures the heart beat rate and pulsatile flow. The changes in optical transmission due to dilation and contraction of vessels are recorded by photoplethysmography (PPG).

Plethysmography assists in recording the changes in tissue volume or organ. For instance, changes in blood volume, lung volume are used to examine the cardiac function and pulmonary function. The changes are measured using an optical light source and a detector. With the wearable sensors, PPG measures the changes in blood volume due to the dilation and contraction of vessels in the dermis and hypodermis area [33–35]. The PPG signal provides deep insight in cardiovascular function such as blood oxygen saturation level, heart rate, stroke volume and vascular resistance [36]. It measures the infrared source for optical imaging about 700–1000 nm and it is the wavelength range for the absorption level of melanin in the skin, protein level.

The absorbance level of oxygenated hemoglobin and un-oxygenated hemoglobin ranges from 600 to 800 nm in pulse oximeter measure oxygen saturation. Several PPG sensors are incorporated into wrist-worn devices that enable wireless transmission. The main advantage of wrist-worn devices is that transmission electronics and battery can be easily attached to strap. The sensors can also be fixed in ear lobe which

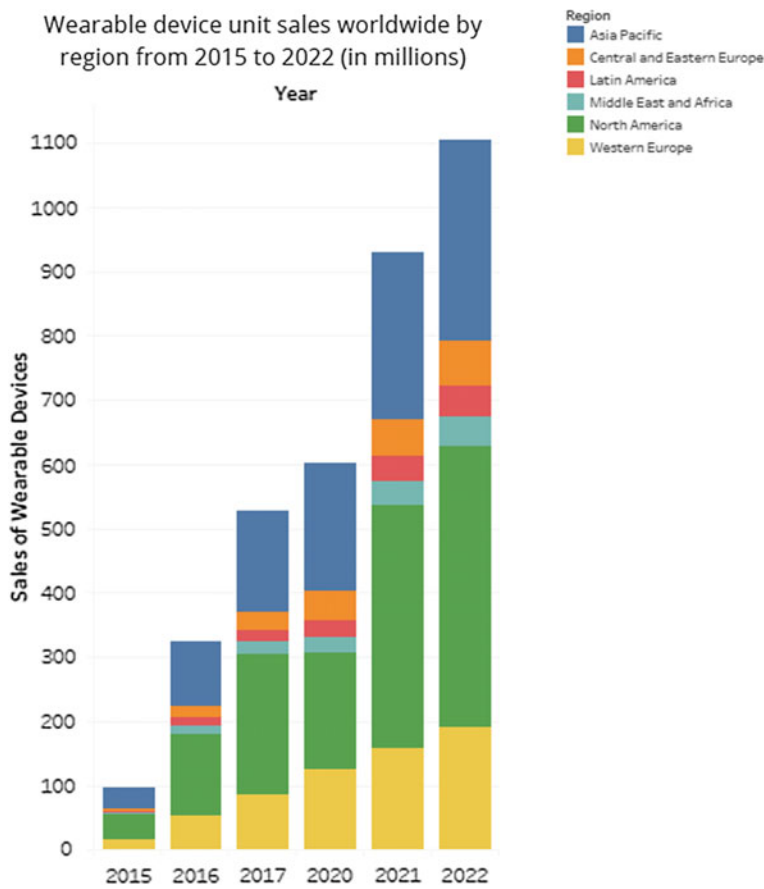


Fig. 10.7 Wearable device unit sales worldwide

increases the signal-to-noise ratio with the high density of vasculature, although eye glass frames or headphones are required to mount the sensor. The disadvantage is the decrease in temperature at the ear lobe that results in peripheral vasoconstriction with decreased signal-to-noise ratio. Some of the characteristics of wearable devices are given in Table 10.1.

10.2 Cloud Computing for WIoT

Cloud Computing is a new emerging platform for data storage and processing. Particular in wearable devices, the storage of data gathered by sensors of these devices are stored on remote cloud through Internet. The popular wearable products include

Table 10.1 Wearable devices characteristics

Accessories	Description	Available prototypes
Smart band	Wrist-worn devices track fitness and other functionalities, without a touchscreen display	Wrist-worn smoking gesture detector, ultrasonic-speaker embedded wrist piece and neck piece, wrist-worn bioimpedance sensor
Wrist watch	Wrist-worn devices with a touchscreen display	Finger-writing with smart-watch, smart-watch life saver
Smart jewellery	Smart jewellery designed for health-monitoring	Typingring, gesture detection ring
Smart foot-wear	Socks, shoes, gloves, or insoles, embedded with sensors	Gait analysis foot worns, foot-worn inertial sensors, LookUp
Smart garment	Clothing items such as shirts, pants, and undergarments serve as wearables	Dopplesleep, Myovibe
Smart eyewear	Spectacles or contact lenses used as wearables	Chroma, iShadow mobile gaze tracker, Google Glass, Google contact lens, indoor landmark identification, object modelling eye-wear, supporting wearables
Sensor patch	Sensor patches could be adhered to the human body skin to track the fitness	DuoSkin, smart tooth patch, tattoo-based iontophoretic-biosensing system

Sony Smart watch, Apple watch, and Fitbit. In case of health care wearable devices, the Cloud allows accessing of health data about the users from anywhere and any-time, by legitimate parties, healthcare professionals, caregivers and patients. The advantages of cloud computing based wearable data are accessibility, connectivity, cost and time efficient, increase in efficient and productivity, scalability to meet the organization requirements, and user centric system services.

The generic architecture of cloud computing based wearable IoT system that include four layers namely wearable device front end system, communication system, cloud support system and extended user applications as depicted in Fig. 10.8. The hardware components include IoT based sensors, actuators, embedded systems and communication handling modules. The software part includes the various application user interface modules and service handling functionalities. The next layers is the wearable communication system layer that constitute of communication gateways such wireless access point and base station for mobile communications.

The objective of communication system is to gather data from sensors present at front systems and then transmit the gathered data to the upper layer cloud support systems. The cloud support system performs the data pre-processing, data analysis, knowledge generation and decision making services according to the user preferences and requirements. The services of the wearable IoT systems are provided to the user and extended users through application interfaces available at the top layer. The

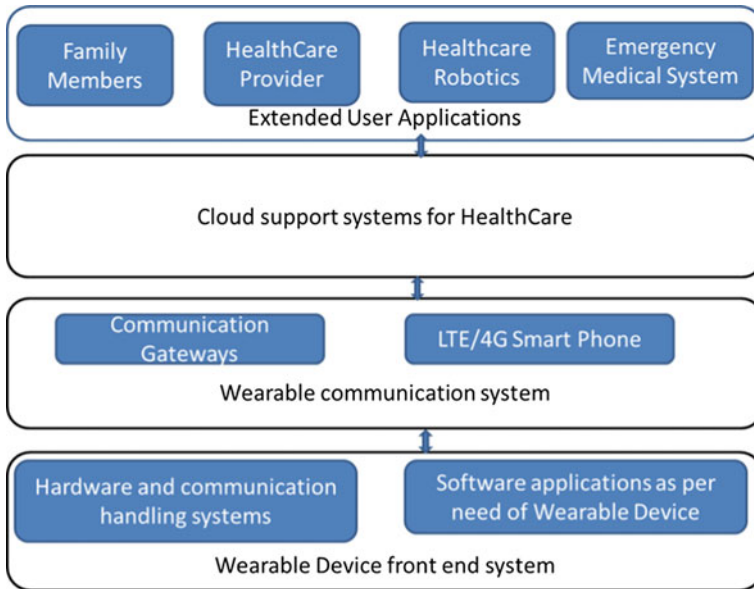


Fig. 10.8 Generic architecture for cloud computing based wearable IoT devices

extended users of wearable devices include family members, healthcare providers, healthcare robotics and any emergency systems. This session will address the various frameworks and mechanisms that proposed in the exiting literature.

10.2.1 Access Control Based Model

Authors Bhatt et al. [37] discussed the access control based framework for wearable IoT using cloud environment. In the wearable IoT domain the access control architecture consists of four layers namely object layer, virtual object layer, cloud services layer and application layer and two primary access control planes as depicted in Fig. 10.9. The object layer consists of the heterogeneous autonomous wearable IoT devices. The functions of these heterogeneous devices are to sense the environment, send to gateways for storage in the cloud, in order to perform data analysis knowledge generation out of the sensed data, and support decision making. The virtual object layer consists of functionalities that map the physical objects underlying in lower layer to the virtual ones. In such case, the upper application layers can perform interaction through virtual objects rather than with physical objects itself of wearable IoT.

The primary objective of these virtual objects is to maintain the current state of participating physical objects. In such case, even when the physical object goes

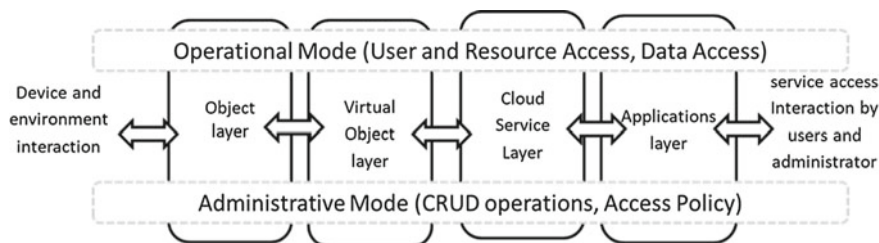


Fig. 10.9 Access control model for cloud based wearable device

offline and the status of these objects are available offline at virtual objects. Once, these physical objects become online, then the status of these physical objects are restored based on the interaction between virtual objects and physical objects. The cloud service layer objective is to provide services such as infrastructure, platform and software. As the physical objects are limited and constrained by resources such as memory, networking, battery, computation power, it is essential to offline the data and tasks of physical object to the cloud environment.

The two access control planes are operational mode and administrative mode. The operational mode controls usage of various systems resources and services by the wearable device users. The objective is to provide users with appropriate permission access to the data, interfaces and services. The administrative mode provides control for the administrator of the wearable system to perform create, read, update and delete various policies and users.

Further, to perform the data analyses on these sensed wearable IoT data input from physical objects, the data mining and machine learning mechanism are supported by these cloud platforms. The top layer is the application layer, where the objective to provide Wearable IoT users with applications essential to meet out their daily needs and demands. These applications provide user friendly interface through which the user can transmit the commands to the application service provider and receive their required responses as application services. Further, the application layers provide support for visualization of data analysed at different service levels.

10.2.2 *BodyCloud*

In context to the resemblance of two technologies namely wearable devices and smart mobile phone, the data generated, such as medical data, by these devices are going to be enormous in volume at unpredicted rate. It is very essential to extract relevant knowledge out these data as important as gathering data from the human body. In essence, the user's role is to wear the sensors on their body; however the underlying processing algorithm should effectively process the knowledge out of the data and offer the users at the minimum cost and time and at quality of service.

According to Fortino et al. [38], there are several ways the cloud computing approach provides infrastructure as service to facility the wearable device data management. Further, such cloud based infrastructure will provide additional functionalities like big data analytics, data mining and machine learning. Authors discussed the BodyCloud (BC) that incorporates Software as a Service (SaaS) approach to support body area network and deployment of cloud-assisted body area network applications.

In this, the cardiac signals are streamed to cloud for further analysis and knowledge generation. The multi-tier architecture of Body cloud constitutes architecture for various applications that integrates body area network and cloud computing platform. Further, this paper proposed data streaming through middle ware and platform for abstract programming, to facilitate the application development for body area network with cloud support systems.

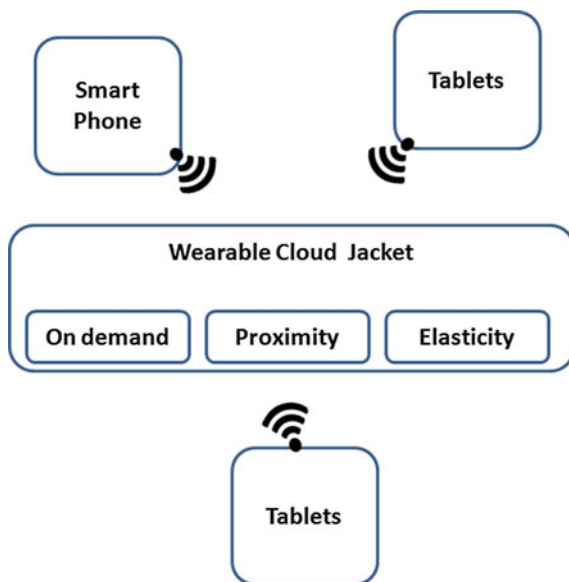
10.2.3 Wearable Glass

Authors in Hao and Helo [39] discussed about wearable glass based on Cloud Manufacturing (CM). In this, the amalgamation of different technologies like Wearable device, cloud computing, IoT, augmented reality play vital role in monitoring and maintenance of industrial manufacturing factories. The primary objective of such architecture is to minimize the manufacturing waste and maximize sustainability. The study addressed how the workers tasks in factory are enhanced through wearable devices.

As stated, the enhancement of human computer interaction is achieved through cloud services and wearable devices and other communication technologies, provided business strategy in terms of demands, services and customer satisfaction. These wearable devices generate enormous amount of data, needed to store on cloud for further processing, analysing and decision making. The primary objective is creating platform for human machine interaction and value enhancement through cooperation among industrial workers. Further, to automate and provide flexible services through cloud, and thus minimizing the workers daily life complexity in manufacturing processes.

The cloud assisted wearable glasses applications are provided with four types of services namely Augmented Reality (AR) service, Infrastructure as a Service (IaaS), AR platform as a service (AR-PaaS) and field services support. The augmented reality service handles the virtual 2D and 3D contents as per the requirements of the users. Each user of the wearable devices are provided with various functional operations like touch, scroll, navigate etc. are handle by these AR service module. Primarily the data gather by different sensors of the Wearable glass is stored at cloud. Next, the Infrastructure as a service is provided to handle difference components like GPS, Camera, microphone, headphone, displays, and sensors as part of the wearable device. Next, AR platform as a service, provide user management functionalities, object recognition, information analysis and visualization and system information retrieving.

Fig. 10.10 Wearable cloud framework



10.2.4 Wearable Cloud

In this modern day, to meet high expectation of consumer, there is requirement of enhanced processors and applications, which further leads to complex and expensive design involved in wearable devices and mobile devices. Authors in Hasan et al. [40] proposed framework to provide cloud services through wearable cloud. The framework for the wearable cloud is depicted in Fig. 10.10. The prototype consists of jacket with minimal computational power, with storage memory capacity of 120 GB. Wearable cloud framework is depicted in Fig. 10.10.

The primary components involved are 10 cost effective Raspberry Pi boards, touch screen interface and winter jacket. To over the resource constraint due to on demand smart phones, and wearable devices, the data gathered from them are stored over the wearable cloud which is connected through communication protocols such as Bluetooth, Wi-Fi and ZigBee to form private cloud. Also, in addition, user can carry the wearable cloud and also get connected to the remote cloud servers through Internet. The major advantages are to provide ad hoc created private cloud, among devices within proximity, and to provide resources to several user devices.

Authors Chen et al. [41] presented the wearable 2.0 framework that focused on the integration of wearable device, health care system and cloud. The work aimed to enhance the Quality of experience (QoE) and Quality of Service (Qos) of user lifestyle. The test bed consisted of connecting smart cloth, and a humanoid robot, a smart phone, and cloud based big data health monitoring system. Openstack framework was used to implement the health cloud services. The functions of health cloud include integrating smart clothing sensors to gather the physiological data and also to

set, monitor the various parameters for sensory devices and communication channel requirements.

10.3 Fog Computing for Wearable IoT Device

Fog computing is a emerging technology targets to bring various services in proximity with users. The services include computation capability, storage, networking, controlling of data processing and user applications services. Particularly, in case of wearable devices, the fog computing intends to bring the computing capability closer to the wearable IoT sensors. This section addresses the various fog computing frameworks and methodologies that exist in the literature.

Seneviratne et al. [42] discussed the need for Fog based computing to meet the resource constraints of wearable devices. Particularly, in case of wearable devices participating in personal area network (PAN) has to minimize the computation power involved in performing intended tasks, computation intensive security overheads, and storage space for data. To handle these constraints, the traditional way of remote offloading of file into data centre or cloud over the Internet is preferred. However, the remote file offloading is not so effective in case of wearable device due to its limited networking capability and resources. Hence, the fog computing is preferable for the reasons that efficient network speed and bandwidth possible in case private PAN of Wearable Devices. Thus, in this way the storage space and file sharing can be effectively handled.

According to Charalampidis et al. [43] discussed the primary objective of fog computing such as storing, pre-processing the enormous amount of data collected from diverse devices, at real-time, enhancing security and privacy, and anomaly detection. In this, the authors proposed Management-as-a-Service (MaaS) for collecting the sensory data from IoT based wearable devices. Also they proposed metrics for the data management services and how to improve the efficiency of wearable device based on IoT platform through failure prediction and detections.

Park and Park [44] argued that enhancement in wearable device computing must be through fog computing by promoting own storage systems, communication capability and computation power. The authors' proposed combinatorial based (k, n) file sharing scheme on fog computing, such that the efficiency of wearable device were improved. The scheme targets on the file storage mechanism and data retrieval from the storage. In the (k, n) combinatorial file sharing scheme, the parameter n represents the number of wearable devices participating in PAN and the parameter k represents the number of devices participating in the data retrieval from the fog storage system. The smart phone was used as the fog node, for the PAN of n wearable devices.

The fog network consists of three layers namely wearable device layer, fog layer and internet layer. The wearable device layer consists of computation constrained devices like smart watch, phone, smart glass, smart bands etc. The fog layer consists of device with much more capabilities than lower layer wearable devices. The fog nodes are aware of the characteristics of the underlying wearable devices and also

the network capabilities in terms of speed and bandwidth of each device. Then the upper layer is the internet layer where, the fog nodes may permanent store the data for further analysis and management in the remote cloud systems.

Borthakur et al. [45] presented the fog computing based framework to perform clustering analytics in health care wearable IoT devices. The paper targeted towards optimizing the computation power by using machine learning at the fog nodes at the proximity of PAN of wearable devices instead of the devices itself. Unlike, the conventional method of cloud based data processing and analysis using machine learning, this authors presented fog based architecture. Under this architecture, the machine learning was applied for the big data gathered from sensors of the healthcare wearable devices. In this case, pathological speech data obtained from audio sensors of tele-monitoring device worn by Parkinson Disease (PD) diagnosed patient. The feature extraction was performed on 164 samples of speech record from Patients with PD.

Name et al. [46] stated that distributed system of computing resources infrastructure enhance the capability of IoT. The authors argued that fog computing as part of distributed computing system that provides support for ubiquitous solution for devices. In this, they proposed Seamless Handover Scheduling policies for Mobile based IPv6 through networking functional modules presented at the fog computing layer. The paper addressed mitigation for mobility issues in smart phone IoT devices.

Taneja and Davy [47] presented the amalgamation of fog and cloud computing. This combined platform provides resource aware offloading of IoT application modules into fog-cloud computing environment.

10.4 Big Data for Wearable IoT

Mezghani et al. [48], the primary applications of wearable devices include the patient health monitoring, analyse human behaviour, vital signals and for therapy management. In conventional method, well define algorithm and data structures are used within confined limitations produces the best possible solutions. However, the recent advancement of wearable healthcare systems involves several dynamic attributes depending on the real time demand of patients.

Moreover, the wearable data sources are from different devices and sensors. These data are of different formats depending on the device. Further, different algorithms are used to encode and transmit these data. Thus, challenge for wearable devices is to handle the scalability and data heterogeneity among the available data and to perform data analysis accurately. Big data provides promising solution to handle the variety, volume and velocity of wearable device data. Further, the authors proposed semantic based big data architecture. The objective of the proposed system was to handle the heterogeneity of the wearable devices and scalability of the big data system through Wearable Knowledge as a Service (KaaS).

Authors in Zhou et al. [49] argued about need for data gathering to provide insight of current user state in wearable healthcare systems. Generally, the body worn sen-

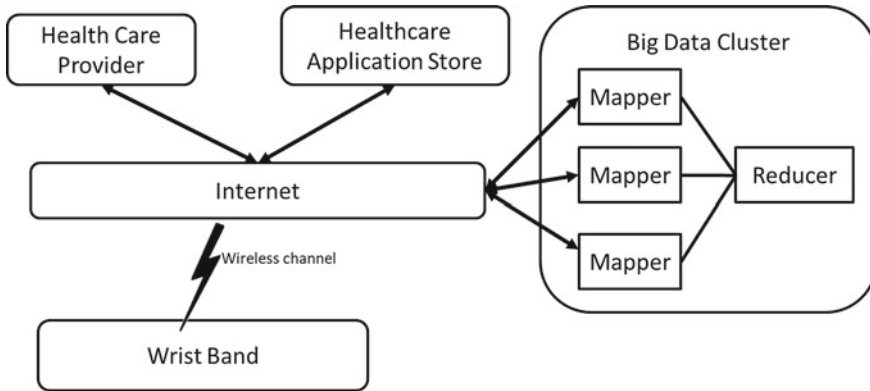


Fig. 10.11 System architecture for wearable health care system based on big data

sensor nodes provide the bio sensed data from patients, this process further involves data collection, storing, and process to identify any health issues related to the users. It is to be noted that, these systems exhibit low level of computing capability and time expensive. In addition, cost heavy throughput in terms of server side computation. Hence, the analysis of raw data of sensory data at real time requires effective mechanism for data handling.

The data produced by such wearable healthcare systems exhibit properties of volume, variety and velocity, and often are unstructured or semi structured data. Hence, Big Data is the effective way to handle these continuous gathering, monitoring and analysis of data. Lin et al. [50] proposed big data architecture for emotional health care monitoring through wearable devices. The authors argued that the most of the existing solution handle the logic reasoning of data gathered from various healthcare systems, and huge gap in emotion aspect of users.

Jiang et al. [51] presented a prototype for the elderly health care system based on Big Data systems. The authors stated that the big data approach would serve the long term goal of elderly wearable healthcare systems. The major challenges need to be addressed in such big data based wearable system are dynamic in input data, non linear trend of sensor data, and multi variant nature of the sensor data.

To overcome these challenges, the authors proposed cognitive and intelligent Hidden Markov Model (HMM) for data forwarding and to detect behaviour pattern of wearable user. Further, the computation and storage overheads are reduced by transmitting only the efficient features of the sensed data for big data analytics. The system performance enhancement is carried out using hashing technique. The proposed prototype consisted of three major components namely a wrist band, a smart phone and a big data system as given in Fig. 10.11.

10.5 Security and Privacy in WIoT

The generic requirements of security and privacy in Wearable IoT remain almost same as other wireless transmission. However, the scale of possible security attacks and communication breach are much more serious issue. The real challenge is the diversity of the wearable devices and its functional wireless communication and network protocols. Moreover, these wearable devices are constrained by resources such as memory, computation, storage, control and networking capability. This section addresses the various security threats in the wearable IoT devices, and various security solutions that exist in the literature.

10.5.1 Attacks in Wearable IoT Devices

- **Physical attack** is intended about the attack on physical components or communication link of the wearable devices.
- **Tampering** of wearable devices is intended to alter or access the sensitive data and information inducing the malfunctioning of the devices.
- **Jamming** is intended to block the wireless communication between transmitter and receiver using a specialized device for jamming purpose.
- **Interference attack** is intended to introduce noise or unwanted signals into the legitimate user's radio frequency communication among wearable devices.
- **Malicious node injection** is intended to add a illegitimate and malicious node into the edge layer of the IoT with the objective of altering the important data and also access the information exchange between the victim wearable devices.
- **Physical damage** is intended to destroy the working components of the wearable IoT devices either partially or completely, so that the legitimate user cannot involve in any sort communication or computation.
- **Malicious code injection** is intended to alter the controlling software and functional components of software framework of wearable devices. In this is highly possible that attack can take complete or partial control of the software components of wearable devices.
- **Sleep deprived attack** is intended to exhaust the battery power of the wearable devices. In this the attack manipulates to increase the energy consumption by involving the victim wearable devices in unwanted computation and execution of software.
- **Network attack** is intended to exploit the networking capability of wearable devices in a communication network.
- **Traffic analysis attack** is intended to sniffing the traffic flow of data in and out of the wearable IoT device and hence injects vulnerability into the traffic.
- **Man-in-the-middle attack** is intended to impose as the legitimate user and involve in malicious network activities among legitimate users, by altering or destroying the valuable communication information of wearable device.

- **Denial of Service (DoS)** attack is intended to flood the traffic network of legitimate participating wearable devices in the network. Due to such malicious attack, the authenticate users are unable to participate in the network communication.
- **Message replaying attack** is intended to confine the legitimate user commands, and manipulate to execute those commands at the attacker convenience.
- **Sybil attack** is to steal the authentication of legitimate user and misuse to access the sensitive data of legitimate user.
- **Software attacks** are intended to inject malware into the software components of the wearable device and attack the normal functioning of these devices.
- **Virus and worms** are malicious code that modify partially or completely the executable codes of the software firmware of wearable devices.
- **Malicious scripts** are intended to perform suspicious execution of code without the knowledge of the legitimate user of the wearable devices.
- **DoS attacks** are intended to maliciously performing operation such that specific importance software services are blocked which are offered by underlying firmware to authentic users.
- **Encryption attacks** are intended to modify the encoding and decoding keys of information exchange through malicious encryption techniques.
- **Side-channel attacks** are intended to exploit the time involved in verification or signing of cryptographic mechanism.
- **Cryptanalysis attacks** are intended to intercepting the keys of encryption by siphoning plain text or cipher texts exchanges by authentic users of wearable devices.

In most of the cases, the wearable devices are proprietary products and licensed software and APIs. Thus, the major challenge in wearable devices is that, there exists no common solution strategy for security and privacy for wearable devices. The research study and literatures targeted universalize of security vulnerabilities through explore, illustrate and classify variety of wearable device. Authors provided extensive summary of the security in data communication for wearable body area network.

The three common such repositories are Common Weakness Enumeration (CVE), Common Vulnerability Exposures (CVE) and National Vulnerability Database (NVD). These vulnerabilities are then mapped according to the specification of devices. These types of classification provide facility to analyse the patterns of attack vulnerabilities and bug. Thus proactive measure to prevent such vulnerability through software patches as per the wearable device applications.

The BLE based communication through Wearable networks encounters three major security and privacy issues namely man-in-the-middle attacks, illegitimate users, and access polices vulnerability the vulnerability of BLE sniffing using Uber-tooth One open software, which works by two mechanism such as eavesdrop through passive way and then inject data packets into the communication channel of BLE through active way.

10.5.2 Conventional Algorithms

In Ometov et al. [52], the authors presented the comparative analysis between conventional security mechanism with the advanced pairing based algorithms for wearable devices. It is to be noted that the conventional security mechanism targets two objectives namely to provide user authentication and then to provide protection during the data transmission along the network link. Generally asymmetric public key cryptography mechanisms such as digital signatures are common for user authentication.

While, the cryptographic algorithms such as RSA, SHA1, SHA2 and AES are commonly used for data protection. These standardized conventional security algorithms follows the principle of confidentiality, integrity and authenticity. These conventional algorithms are based on mathematically modelling, discrete logarithm, and integer factorization solution. The mathematical primitives applied on cryptographic calculation are based on big integer that involves operations like multiplication, division, exponential functions and elliptic curve mechanisms.

10.5.2.1 Bilinear Pairing Operation

On the other hand, authors discussed how to enhance the privacy mechanisms by the bilinear pairing algorithm. Under bilinear pairing algorithm, the various schemes are possible namely user identity based data encryption, anonymous credentials through key value pair attributes, and group based signature mechanism. In case of wearable IoT devices, dynamic accumulator mechanism is used to revocation of invalid device entry into communication and thus facilitating the attacker identification efficiently.

The generalized function for bilinear pairing operations considers two group of elements G_A and G_B , which are mapped to each other through mapping function m to the cryptographic output group G_M given by Eq. 10.1 below.

$$m = G_M \leftarrow G_A \times G_B \quad (10.1)$$

The computation of mapping function m is based on Miyaji—Nakabayashi—Takano (MNT) Curves, Elliptical cryptographic Curves (ECC), and Barreto—Naehrig (BN) curves. There are two types of pairings are possible namely asymmetric and symmetric methods. In asymmetric pairing methods, the two input groups are asymmetric in nature $G_A \neq G_B$, on the other hand in symmetric pairing method; the two input groups are symmetric in nature $G_A = G_B$. The examples of such pairing algorithms are Tate, Ate, Eta, O-Ate, and Weil.

Further authors in Ometov et al. [52] conducted experimentation based on the pairing based cryptographic mechanism on several wearable devices such as Apple smart watch, Sony Smart watch, Intel Edison systems, Raspberry Pi Devices, Samsung Galaxy smart phone and Apple iPhone 6. The experimental results provided insight into the need for further enhancement in cryptographic solution for wearable devices, also comprehensive report on comparison between conventional crypto-

graphic algorithms versus pairing based cryptographic algorithms for wearable IoT devices.

10.5.2.2 Data Driven Security (SIFT Method)

Cai et al. [53] proposed Signal Feature Correlation based Testing (SIFT) method for the identification of breach on medical data such as Electro Cardiogram (ECG) signal and Artery Blood Pressure (ABP) signal measurements fetched from wearable health care IoT device of a patient. Basically, SIFT method provides data driven security measure for wearable devices. This SIFT method exploits the common correlation between two different physiological signals obtain through same physical monitoring process.

The ECG signal and ABP signal are difference in their physical properties but meant for cardiac physiological process. Hence, using intelligent mechanisms, the ECG signal features are correlated with ABP signal features, traced over the period of time series. To perform the identification of interrelationship between these two signals, the machine learning algorithms are used. The features are extracted from each of these signals. Then the features are mapped among each other, and are trained through support vector machine (SVM) prediction model. This correlation is effectively used to detect any breach in sensor device and being misused by unauthorized users or hijacked.

This mechanism of SIFT method is powerful in detecting multiple users of sensors in wearable IoT. As, the SIFT works on the principle of single level of redundancy, any hijacking attacks of wearable sensors can be detected effectively. The SIFT methods has three phases namely feature extraction phase, training phase and detection phase. During the feature extraction phase, the ECG signal $E(t)$ and the ABP signal $A(t)$ are recorded at time t . Then these are normalized over the period of time ω where t lies within the range of $[0, \omega]$. The portrait of features is obtained using the function $P(t) \leftarrow A(t), E(t)$.

Then, from the $P(t)$, salient features are extracted. The features extracted are categorized further into geometric features and matrix features. The geometric feature corresponds to the absolute values for example Systolic blood pressure peak in ABP signals, R peak amplitude in ECG signals. The matrix feature represents the count of points that actually belongs to the portrait function, the standard deviation and average of the feature matrix. Next, during the training phase, the suitable machine learning model is established for specific user of the Wearable IoT. The common window of Δ time unit is used to collect ECG and ABP signals for the users.

Two type of class are used to represent the validation of signal correlation namely positive and negative correlations. The positive correlation among different features of ECG and ABP signal indicates the authenticate user of the sensors. While the negative correlation among signal features indicates intruder or sensors being hijacked. Finally, in the detection phase, the testing data from individual monitored over ω time period is fed into the SVM machine learning model for classification purpose.

10.5.2.3 Gait Based Authentication

The objective of wearable devices is to enhance the quality of life and improve daily activities of humans by physical proximity of the technology. However, these wearable devices are limited in terms of computation, storage, battery life and security aspects. Sun et al. [54] proposed human gait acceleration based wearable device authentication mechanism. The walking style of human is a kind of unique biometric parameter that can be used to measure the acceleration factor associated with each wearable user. The inbuilt accelerometer within the wearable device is considered for collecting acceleration signals. The authors presented that the proposed authentication system as cost effective solution, without any addition hardware interaction with the users.

10.5.2.4 Handshaking Based Secured Channel Communication

Shen et al. [55] proposed handshaking mechanism between wearable devices used to perform secured data exchange. In this method, named as Shake-n-Shack, the physical handshaking motion of two different user utilized for device authentication between two wearable devices of users within contact. The hand shaking pattern of two wearable user generate a unique motion based on which symmetric key is generated to authenticate secured communication between the wearable devices.

The advantages of this method are user friendly, not need for complex hardware, no predefined gestures mechanisms of the users. The major components of the proposed system are handshake motion detection, feature extraction, key quantization and reconciliation, wireless communication configuration, and data encryption/decryption. In handshake motion detection module, Inertial Measurement Units (IMU) captures the signal generated during handshaking between the wearable users. This raw inertia signals are then gathered for processing the signal feature extraction.

Next, in the feature extraction modules, based on Principle Component Analysis (PCA) methods, the signal bit rate is estimated. The various performance metrics namely bit rate, bit agreement rate, coherence of signals, acceptance rate, rejection rate, error rate, time and energy consumption are measured. Next, the key quantization and reconciliation module, reduces the generated bit rate by discarding the ambiguous bits from original signals.

For the given mean (μ), standard deviation (σ) and quantization parameter (Q), the user key bit position and values are determined. In key reconciliation process, keys generated by two wearable users (K_A & K_B) in action are verified for equivalent i.e., $K_A = K_B$ and hence agreed for secure communication channel. In wireless channel configuration, based on the candidate key agreement, the configuration of wireless channel is set up through probing mechanism and then the secured data exchange is initiated. Table 10.2 summaries the various security measures and solutions that exists in the literature:

Table 10.2 Summary of various security solutions for wearable devices

Objective	Authors	Wearable device	Method
Data security	Ometov et al. [52]	Smart watch	Pairing based Cryptography
Data security	Cai et al. [53]	Amulet	Signal Feature Correlation based Testing (SIFT)
Data security	Shen et al. [55]	Wrist worn smart watch	Handshaking based symmetric key generation
File sharing	Park and Park [44]	Wearable PAN device	Combinatorial (k, n) scheme
User device authentication	Sun et al. [54]	H Shirt	Gait based
Data security & privacy	Usman et al. [56]	Wireless body area network	Key management & device authentication

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